

# MODULE 5 — Campaign Management & Orchestration (Unified Omni-Channel Brain)

**Version:** 1.0

**Status:** Draft → Pending Confirmation

**System:** Omni-Channel Martech CRM Platform

**Audience:** Product, Backend, AI, Growth, QA, Compliance

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## 1. Module Overview & Strategic Role

### 1.1 Purpose

This module is the **central orchestration engine** that coordinates **who to contact, when, through which channel, with what logic, and at what cost.**

If:

- Module 3 = data truth
- Module 4 = execution power
- Module 4A/B/C = channels

Then **Module 5 is the brain.**

It ensures:

- Consistent customer experience
  - Cost-disciplined execution
  - Cross-channel intelligence
  - Measurable outcomes
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## 2. Scope

### 2.1 In Scope

- Campaign creation & configuration
- Omni-channel orchestration (Call, WhatsApp, SMS, Email)
- Audience segmentation
- Trigger-based and scheduled execution

- AI-driven branching logic
- Budget & usage control
- Campaign lifecycle management

## **2.2 Explicitly Out of Scope**

- Channel-specific UI (handled in 4A/4B/4C)
  - Billing & invoicing (Module 9)
  - Analytics dashboards (Module 7)
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# **3. Campaign Entity Definition**

## **3.1 Core Campaign Attributes**

Each campaign must include:

- Campaign ID (system generated)
  - Campaign Name
  - Campaign Type
  - Primary Channel
  - Secondary / Fallback Channels
  - Objective (configurable)
  - Status (Draft / Active / Paused / Completed / Archived)
  - Created by (User / System)
  - Start date & time
  - End date (optional)
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## **3.2 Campaign Types (Confirmed)**

Supported campaign archetypes:

- Sales outreach
- Lead qualification
- Follow-up / nurture
- Support notification
- Collections
- Surveys / feedback
- Transactional alerts

Campaign types influence:

- Allowed channels
  - AI behavior
  - Compliance rules
  - Cost limits
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## 4. Audience Selection & Segmentation

### 4.1 Audience Definition

Campaign audience is defined via **CRM filters (Module 3)**:

- Lead source
- Pipeline & stage
- Score range
- Last activity
- Tags
- Assigned owner
- Consent status

Audiences are **dynamic by default** unless explicitly frozen.

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### 4.2 Exclusion Rules

Campaigns must support exclusions:

- Opt-out leads
- DND / quiet hour conflicts
- Recently contacted leads
- Active conversations
- Manual suppression lists

Exclusions are evaluated **at execution time**, not only at campaign start.

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## 5. Omni-Channel Execution Model

### 5.1 Channel Sequencing

Campaigns may define:

- Single-channel execution
- Multi-step channel sequences

Example:

1. AI Call
  2. If no answer → WhatsApp after 2 hours
  3. If no response → SMS next day
  4. If engaged → Email follow-up
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## 5.2 Channel Selection Rules

Channel choice may depend on:

- Lead preference
- Language
- Timezone
- Cost thresholds
- Previous engagement history

Rules must be deterministic and logged.

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# 6. AI-Driven Campaign Logic

## 6.1 AI Decision Capabilities

AI may:

- Decide next best channel
- Choose message variant
- Adjust timing
- Escalate to human
- Terminate campaign journey early

AI **cannot**:

- Override compliance rules
  - Exceed campaign budget
  - Ignore opt-out signals
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## 6.2 AI Inputs

AI decisions are based on:

- Lead profile & history
  - Previous campaign outcomes
  - Channel performance
  - Time & context
  - Tenant-defined guardrails
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# 7. Campaign Flow Designer (Hybrid)

## 7.1 Rule-Based Configuration

Supports:

- IF / ELSE conditions
- Delays
- Counters
- Retry limits

Used for:

- Simple campaigns
  - Predictable flows
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## 7.2 Flow-Based Orchestration

Node-based flow builder with:

- Channel nodes (Call / WhatsApp / SMS / Email)
- Decision nodes
- Wait timers
- Exit conditions
- Error handling paths

Used for:

- Complex journeys
- Enterprise automation
- Compliance-heavy workflows

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## 8. Execution Triggers

Campaigns may start based on:

- Manual trigger
- Scheduled time
- Lead creation
- Stage change
- AI call outcome
- Inactivity timeout
- External webhook (future)

Triggers must be logged with source attribution.

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## 9. Cost Control & Budget Management (Critical)

### 9.1 Budget Definitions

Campaigns must support:

- Total budget cap
  - Daily spend limit
  - Per-lead spend limit
  - Channel-specific caps
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### 9.2 Enforcement Rules

When limits are reached:

- Campaign auto-pauses
- Pending actions cancelled
- Admin notified

No soft overages allowed unless explicitly approved.

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## 10. Campaign State Management

## 10.1 Campaign States

- Draft
- Active
- Paused
- Completed
- Archived

State transitions must be:

- Logged
  - Attributable
  - Non-destructive
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## 10.2 Lead-Level Campaign State

Each lead maintains:

- Current campaign step
  - Last action
  - Next scheduled action
  - Exit reason (if exited)
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# 11. Human Intervention & Overrides

## 11.1 Allowed Overrides

Authorized users may:

- Pause campaign
- Skip lead step
- Reassign lead
- Force channel change
- Escalate to human

Overrides must:

- Require permission
- Be logged
- Not break compliance rules

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## 12. Failure Handling & Recovery

### 12.1 Failure Scenarios

- Channel delivery failure
- AI execution failure
- Time window violation
- Budget exhaustion

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### 12.2 Recovery Actions

- Retry (configurable)
- Switch channel
- Exit campaign
- Schedule follow-up
- Flag for review

Failure logic must be campaign-specific.

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## 13. Activity Timeline & Audit Integration

Every campaign action must be written to:

- Lead Timeline (Module 3)
- Campaign execution log
- AI decision log

Logs are:

- Immutable
  - Time-stamped
  - Filterable
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## 14. Permissions & Security



- Campaign creation restricted by role
  - Execution permissions configurable
  - No cross-tenant campaign visibility
  - RBAC enforced via Module 2
  - License limits enforced via Module 1
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## 15. Data Outputs (For Analytics Module)

This module emits:

- Campaign performance events
- Channel conversion markers
- Cost consumption data
- AI vs human effectiveness signals

Visualization is handled in **Module 7**.

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## Closing Position (Straight Talk)

This module determines **whether the platform scales or collapses**.

Without orchestration:

- Channels spam
- Costs leak
- AI behaves randomly

With orchestration:

- Conversations feel intentional
  - Spend stays controlled
  - AI becomes predictable, not scary
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